

JEE MAIN REVISION MASTER: ELECTROMAGNETIC INDUCTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 21 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Faraday-Lenz Law, Induced EMF & Motional Induction	17 PYQs	Electromagnetic Induction
Moderate	Self & Mutual Inductance & Energy Stored	5 PYQs	Inductance

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Faraday-Lenz Law, Induced EMF & Motional Induction	$\mathcal{E} = -N \frac{d\Phi}{dt}$, $\mathcal{E}_{\text{trans}} = B v L$, $\mathcal{E}_{\text{rot}} = \frac{1}{2} B \omega L^2$, $\Delta q = \frac{\Delta \Phi}{R}$
Self & Mutual Inductance & Energy Stored	$L = \frac{N\Phi}{I} = \mu_0 n^2 A l$, $U = \frac{1}{2} L I^2$, $M = k \sqrt{L_1 L_2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Faraday-Lenz Law, Induced EMF & Motional Induction (Very High Yield — 17 PYQs)

Core Principle: Faraday's Law: $\mathcal{E} = -N(d\Phi / dt)$. Lenz's Law dictates that induced effects oppose the cause (energy conservation). Motional EMF for translating rod $\mathcal{E} = B v L$; for rotating rod pivoted at one end $\mathcal{E} = (1 / 2) B \omega L^2$. Induced charge $\Delta q = (\Delta \Phi / R)$ depends only on total flux change, NOT on the rate of change.

■ **Caution / JEE Pitfall:** Charge flown $\Delta q = \Delta \Phi / R$ is INDEPENDENT of time taken! Whether the flux changes in 1 millisecond or 1 hour, the same charge flows.

■ **Speed Trick:** For an aircraft flying horizontally in the northern hemisphere, vertical component of Earth's magnetic field B_v induces an EMF across the wings: $\mathcal{E} = B_v v L$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the figure shown, the conductor PQ of length l is moved from $x=0$ to $x=b$ and then up to $x=2b$ with a constant velocity $\vec{v}$ " [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "When a bar magnet is pushed towards the coil, along its axis, as shown in the figure, the galvanometer pointer deflects towards X . When thi..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "An induced current of 2 A flows through a coil. The resistance of the coil is 10Ω . What is the change in magnetic flux associated wi..." [Ans: N/A]

3.2 Self & Mutual Inductance & Energy Stored (Moderate Yield — 5 PYQs)

Core Principle: Self inductance $\mathcal{E} = -L(dI/dt)$, where $L = \mu_0 n^2 A l$ for a solenoid ($L \propto N^2$). Mutual inductance $\mathcal{E}_2 = -M(dI_1/dt)$, with coupling factor $M = k\sqrt{L_1 L_2}$ ($0 \leq k \leq 1$). Magnetic energy stored in inductor $U = (1/2)LI^2$.

■ **Caution / JEE Pitfall:** Self inductance of a solenoid is proportional to N^2 (square of turns), not N ! If turns are doubled, inductance quadruples.

■ **Speed Trick:** Eddy currents are minimized by using laminated magnetic cores with insulating varnish (used in transformers and chokes).

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "In Faraday-Henry's experiment, a coil is connected to a galvanometer. For the deflection of pointer in the galvanometer, which of the follow..." **[Ans: N/A]**
- **[2024 Shift] Q#2 (MCQ):** "A square loop of side length a is moving away from an infinitely long current carrying conductor at a constant speed v as shown. Let $x...$ " **[Ans: C]**
- **[2022 Shift] Q#2 (MCQ):** "A long solenoid has 500 turns, when a current of 2A is passed through it, the resulting magnetic flux linked with each turn of the solenoi..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the figure shown, the conductor PQ of length l is moved from $x=0$ to $x=b$ and then up to $x=2b$ with a constant velocity $\vec{v}$. A uniform magnetic field $\vec{B}$ is perpendicular to the plane of the paper and extends from $x=0$ to $x=b$ and it is zero from $x>b$. The magnitude of emf Induced in the conductor is

- (A) $Blv; 0 \leq x \leq b$
- (B) Zero; $0 \leq x \leq b$
- (C) $Blv; 0 \leq x \leq b$
- (D) $Blv; b \leq x \leq 2b$

Correct Answer: None

Solution Analysis: When conductor PQ moves from $x=0$ to $x=b$, it is inside the uniform magnetic field $\vec{B}$. The magnetic field is perpendicular to the plane of the paper. The conductor has length l and moves with constant velocity $\vec{v}$. For a straight conductor moving perpendicular to a magnetic field, the motional emf induced across its ends is $\mathcal{E}=Blv$. Here, B , l , and v are constant. So, the induced emf remains constant while the conductor is in the region $0 \leq x \leq b$. When the conductor moves from $x=b$ to $x=2b$, it is outside the magnetic field region. In this region, the magne

Benchmark Problem 2: KCET 2026 (Q#2)

In Faraday-Henry's experiment, a coil is connected to a galvanometer. For the deflection of pointer in the galvanometer, which of the following statement/s is/are WRONG? The pointer in the galvanometer deflects - (a) When the bar magnet is moved towards the stationary coil along its axis. (b) When the bar magnet is moved away from the stationary coil along its axis. (c) When the coil is moved towards the stationary bar magnet along its axis. (d) When the coil and the magnet are moved without relative motion between them.

- (A) a and b
- (B) b and c
- (C) a, b and c
- (D) Only d

Correct Answer: None

Solution Analysis: The wrong statement is (d) . In Faraday-Henry's experiment, the galvanometer pointer deflects only when the magnetic flux linked with the coil changes. When there is relative motion between the magnet and the coil, the magnetic flux through the coil changes. So an induced current is produced, and the galvanometer shows deflection. Now check each case: (a) When the bar magnet is moved towards the stationary coil, the magnetic flux through the coil changes. So the pointer deflects. This statement is correct. (b) When the bar magnet is moved away from the stationary coil, the magnetic flux again